

Aidan Ream

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Overview

Computer Graphics Programmer and Technical Artist passionate about real time rendering, CPU and GPU performance optimizations, shader programming, 3D art workflows, and visual design fundamentals.

Projects

Monte Carlo Ray Tracer

- Made a C++ Path Tracer with the following features: Geometric primitive and triangle-mesh support, performance optimization structures, multi-threading, png output, Importance Sampling, Atmospheric Persepective camera effects, and more.
- Portfolio Link: [here!](#) Tools Used: C++

Compute Shader Based Vulkan Cloth Simluation

- Created a real time simulation environment with the Vulkan graphics library that utilizes a spring based compute shader on a grid of connected points to simulate the movement and flow of cloth using the GPU for performance optimization.
- Portfolio Link: [here!](#) Tools Used: C++, Vulkan

Skeletal Animation System For Custom Game Engine

- Collaborated on a 3D Roguelike adventure game created by building a custom game engine with OpenGL. My primary contributions were implementing a skeletal animation system, 3D modeling, rigging, and animations, and creating the debug mode.
- Portfolio Link: [here!](#) Tools Used: C++, openGL, Blender

Fractal Romanesco Broccoli Raymarching Fragment Shader

- Created a Shader that utilizes raymarching, a graphics technique that steps through a scene defined by signed distance functions, to render a 3D fractal shape in the form of my favorite mathematical vegetable.
- Portfolio Link: [here!](#) Tools Used: C++, openGL, GLSL

Embedded Systems Programming for F.R.I.D.A Animatronic

- The F.R.I.D.A animatronic was a continuation of a project by the Liberal Arts and Engineering Studies Club. Our team implemented an LLM backed conversation system trained on our departments information, Microphone input and Speaker Output, Face Tracking, and Servo Motor Head rotation powered by an internal Raspberry Pi.
- Tools Used: Raspberry Pi, Arduino, Python

Skills

Programming Languages: C++, C, Assembly Language, Java, Python, GLSL, glTF, HLSL, SPIR-V

Graphics Libraries & Game Engines: openGL, Vulkan, Houdini, Unity, Unreal Engine 5, Godot, Nvidia NSight Graphics, RenderDoc

Visual Design Software: Maya, Blender, Substance Painter, Photoshop, Adobe Illustrator

Other Skills: Agile/Scrum workflows, Git source control, Great group communication, Team management

Experience

Calpoly, Lab Manager at Expressive Technology Studios

- Provided software updates and equipment upkeep for lab equipment
- Ran 10+ events per year utilizing our spatial audio theater space

San Luis Obispo, CA
August 2023 – December 2025

ACM SIGGRAPH, Student Volunteer

- Assisted conference attendees with navigation and scheduling
- Knowledgeable about the various disciplines of computer graphics, and where they were located around the conference center.

Vancouver, BC - August 2025
Los Angeles, CA - July 2026

Education

California Polytechnic State University, San Luis Obispo CA

Bachelor of Science in Liberal Arts and Engineering Studies

Computer Graphics concentration - Coursework focused on computer graphics projects

2D and 3D Animation concentration - Coursework focused on design principles and industry standard workflows/technologies

Computer Science Minor - Additional coursework taken exploring various computer science disciplines

Graduated June 2026

Cumulative GPA 3.54